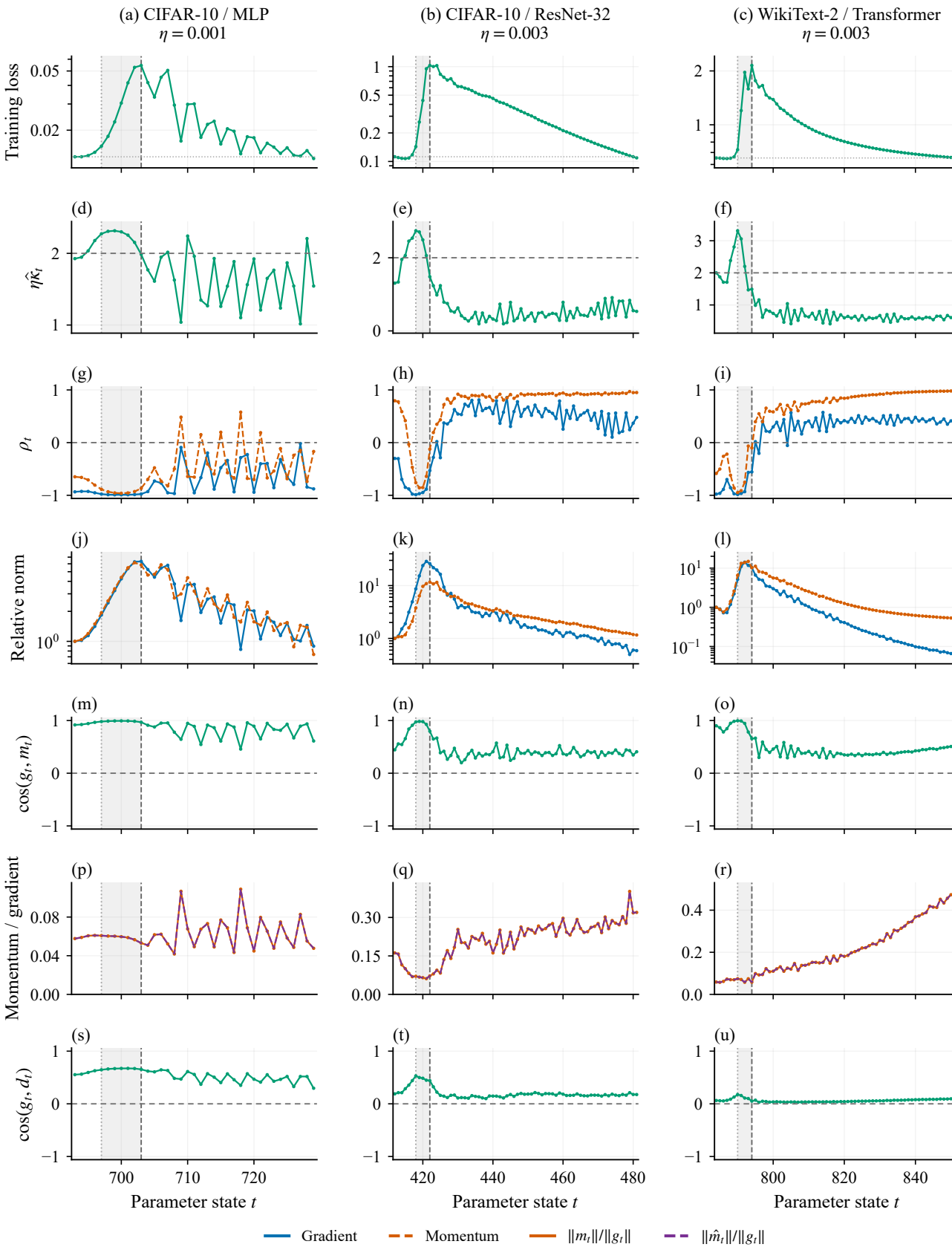

(a) CIFAR-10 / MLP
$\eta = 0.001$
(b) CIFAR-10 / ResNet-32
$\eta = 0.003$
(c) WikiText-2 / Transformer
$\eta = 0.003$
Training loss
(d)
(e)
(f)
$\hat{\eta}\hat{\kappa}_t$
(g)
(h)
(i)
ρ_t
(j)
(k)
(l)
Relative norm
(m)
(n)
(o)
$\cos(g_t, m_t)$
(p)
(q)
(r)
Momentum / gradient
(s)
(t)
(u)
$\cos(g_t, d_t)$
Parameter state t
Parameter state t
Parameter state t
Gradient
Momentum
$\|m_t\|/\|g_t\|$
$\|\hat{m}_t\|/\|g_t\|$